be associated with particular obstacle categories. Subgoals are concerned with the amount of resources to be applied to achieve the primary goals. Boundary conditions are given by the states themselves in combination with obstacle categories, so that each problem has two centers of interest, the associated system state and the cause of that particular state.

Conclusion

In conclusion of this discussion of Problem Formulation the Exploration of Obstacles forms the last half of the step, being preceded by Statement Clarification. While Exploration of Obstacles alone contains the generation and ranking of problems which are the objects of the Problem Formulation step of a system study, it relies basically upon a clear interpretation of the Problem Statement as a system. It is during Statement Clarification that the Problem Statement is reworked from an imprecise but intuitive and idealized description of a goal into a formal but still rather abstract definition of a system. In this process, some concepts of system theory and symbolic logic are useful tools: the notion of a statement or sentence as a proposition, the requirements of a statement which defines a system, and the idea of the states of a system. As a final comment, it should be noted that Exploration of Obstacles requires methods for determining and scaling need for various operational features in the system, and suitable theoretical techniques cannot at this time be named. The methodology of Exploration of Obstacles depends, then, on informal reasoning.

References

[1] A. D. Hall, *A Methodology for Systems Engineering*. Princeton, N. J.: Van Nostrand, 1962.
[2] L. von Bertalanffy, "An outline of general system theory," *British J. for the Philosophy of Science*, vol. I, no. 2, August 1950.
[3] D. O. Ellis and F. J. Ludwig, *Systems Philosophy*. New York: Prentice-Hall, 1962.
[4] "Systems, organizations, and interdisciplinary research," in *Systems: Research and Design*, D. P. Eckman, Ed. New York: Wiley 1961, Ch. 2.
[5] F. B. Fitch, *Symbolic Logic—An Introduction*. New York: Ronald Press, 1952.
[6] L. A. Zadeh and C. A. Desoer, *Linear System Theory—The State Space Approach*. New York: McGraw-Hill, 1963.

Information Value Theory

RONALD A. HOWARD, SENIOR MEMBER, IEEE

Abstract—**The information theory developed by Shannon was designed to place a quantitative measure on the amount of information involved in any communication. The early developers stressed that the information measure was dependent only on the probabilistic structure of the communication process. For example, if losing all your assets in the stock market and having whale steak for supper have the same probability, then the information associated with the occurrence of either event is the same. Attempts to apply Shannon's information theory to problems beyond communications have, in the large, come to grief. The failure of these attempts could have been predicted because no theory that involves just the probabilities of outcomes without considering their consequences could possibly be adequate in describing the importance of uncertainty to a decision maker. It is necessary to be concerned not only with the probabilistic nature of the uncertainties that surround us, but also with the economic impact that these uncertainties will have on us.**

In this paper the theory of the value of information that arises from considering jointly the probabilistic and economic factors that affect decisions is discussed and illustrated. It is found that numerical values can be assigned to the elimination or reduction of any uncertainty. Furthermore, it is seen that the joint elimination of the uncertainty about a number of even independent factors in a problem can have a value that differs from the sum of the values of eliminating the uncertainty in each factor separately.

Manuscript received March 1, 1965.
The author is Professor of Engineering-Economic Systems, Stanford University, Stanford, Calif.

Notation

A SPECIAL NOTATION will be used to make as explicit as possible the conditions underlying the assignment of any probability. Thus,

x = a random variable
A = an event
$\mathcal{S}$ = the state of information on which probability assignments will be made
$\{x|\mathcal{S}\}$ = the density function of the random variable x given the state of information
$\{A|\mathcal{S}\}$ = the probability of the event A given the state of information
$\langle x|\mathcal{S}\rangle$ = the expectation of the random variable x, which equals $\int_x x\{x|\mathcal{S}\}$
$\mathcal{E}$ = the experience brought to the problem, the special state of information represented by total a priori knowledge
$\{x|\mathcal{E}\}$ = the density function of a random variable x assigned on the basis of only a priori knowledge $\mathcal{E}$; designated the prior on x.

Our notation does not emphasize the difference in probability assignment to a random variable and to an event because the context always makes clear the appropriate interpretation. We should interpret the generalized summation symbol $\int$ used in the definition of $\langle x|\mathcal{S}\rangle$ as an integral if the random variable is continuous and as a summation if it is discrete.

An Inferential Concept

A most important inferential concept is expansion; it allows us to encode our knowledge in a problem in the most convenient form. The concept of expansion permits us to introduce a new consideration into the problem. Suppose, for example, that we must assign a probability distribution to a random variable u. We may find it much easier to assign a probability distribution to u if we had previously specified the value of another random variable v and we may also find it easy to assign a probability distribution directly to v. In this case the expansion equation,

$$\{u|\mathcal{S}\} = \int_v \{u|v\mathcal{S}\}\{v|\mathcal{S}\} \tag{1}$$

shows that all we have to do to find $\{u|\mathcal{S}\}$ is multiply $\{u|v\mathcal{S}\}$ by $\{v|\mathcal{S}\}$ and sum over all possible values of the random variable v. Mathematically this equation is no more than a consequence of the definition of conditional probability, but in the operational solution of inference problems it is extremely valuable. If we want to find only the expectation of u, $\langle u|\mathcal{S}\rangle$, rather than the entire density function $\{u|\mathcal{S}\}$ we apply expansion in the form,

$$\langle u|\mathcal{S}\rangle = \int_u u\{u|\mathcal{S}\} = \int_v \int_u u\{u|v\mathcal{S}\}\{v|\mathcal{S}\} = \int_v \langle u|v\mathcal{S}\rangle\{v|\mathcal{S}\}. \tag{2}$$

This equation shows that in this case we must only provide the expectation of u conditional on v and the probability distribution of v in order to find the expectation of u.

The inferential concept of expansion allows us to go directly from the statement of an inference problem to its solution in simple logical steps. It provides a link between the formalism of probability theory and the path of human reasoning; it is a tool of thought.

A Bidding Problem

To illustrate our approach we shall consider a specific problem. Suppose that our company is bidding on a contract against a number of competitors. We shall let p be our company's cost of performing on the contract; unfortunately, we are uncertain of this cost. We let ℓ be the lowest bid of our competitors, and as you might expect, this too is uncertain. Our problem is to determine b, our company's bid on the contract. Our objective in this determination is to maximize the expected value of v, our company's profit or value from the contract.

Naturally, our company will not win the contract if its bid b is higher than the lowest bid of our competitors ℓ, thus, in this case our company's profit is zero. However, if our bid b is less than the lowest competitive bid ℓ our company will obtain the contract and will make a profit equal to the difference between our bid b and our cost p. Thus, profit v to our company is defined by

$$v = \begin{cases} b - p & \text{if } b < \ell \\ 0 & \text{if } b > \ell \end{cases}. \tag{3}$$

This equation indicates that the probability distribution of profit given our bid $\{v|b\mathcal{E}\}$ would be trivial to determine if we only knew our company's cost p and the lowest bid of our competitors ℓ, since by the expansion concept

$$\{v|b\mathcal{E}\} = \int_{p,\ell} \{v|bp\ell\mathcal{E}\}\{p,\ell|b\mathcal{E}\}. \tag{4}$$

In this equation $\{p,\ell|b\mathcal{E}\}$ represents the joint distribution of our cost and our competitor's lowest bid given our bid and the state of knowledge $\mathcal{E}$ that we brought to the problem. However, since we are interested only in the expected profit by assumption we can write (4) in the expectation form,

$$\langle v|b\mathcal{E}\rangle = \int_{p,\ell} \langle v|bp\ell\mathcal{E}\rangle\{p,\ell|b\mathcal{E}\}. \tag{5}$$

At this point we shall make two assumptions. The first is that our cost p and the lowest competitor's bid ℓ do not depend upon our bid b; that is,

$$\{p,\ell|b\mathcal{E}\} = \{p,\ell|\mathcal{E}\}. \tag{6}$$

The second assumption is that our cost p is independent of the lowest competitive bid ℓ, or,

$$\{p,\ell|\mathcal{E}\} = \{p|\mathcal{E}\}\{\ell|\mathcal{E}\}. \tag{7}$$

With these assumptions (5) becomes

$$\langle v|b\mathcal{E}\rangle = \int_{p,\ell} \langle v|bp\ell\mathcal{E}\rangle\{p|\mathcal{E}\}\{\ell|\mathcal{E}\}. \tag{8}$$

In view of (3) we have immediately that the expectation of profit conditional on our bid, our cost, and the lowest competitive bid is simply

$$\langle v|bp\ell\mathcal{E}\rangle = \begin{cases} b - p & \text{if } b < \ell \\ 0 & \text{if } b > \ell \end{cases}. \tag{9}$$

The next step is to assign prior probability distributions $\{p|\mathcal{E}\}$, $\{\ell|\mathcal{E}\}$ to our cost and to the lowest competitive bid. The probability distribution on our cost $\{p|\mathcal{E}\}$ would be based upon the information that our company had gathered in performing similar contracts as amended by the technical considerations involved in the new contract. The prior distribution on the lowest competitive bid $\{\ell|\mathcal{E}\}$ would be based on our experiences in bidding against our competitors on previous occasions and on other information such as reports in the trade press. Although we shall not have the opportunity to digress on the subject of how these assignments are made, suffice it to say that effective procedures for this purpose are available.

To gain ease of computation we shall state simply that in this problem the prior distribution on our company's cost p is a uniform distribution between zero and one and that our prior distribution on our competitor's lowest bid is a uniform distribution between zero and two. These prior

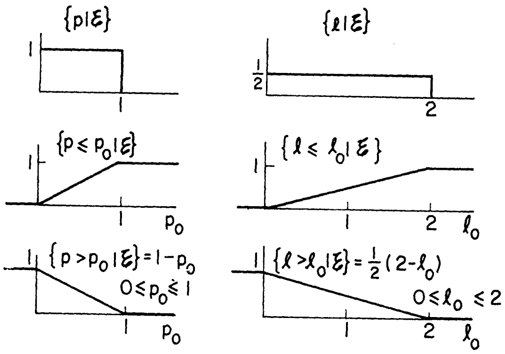

Fig. 1. Priors on cost of performance and lowest competitive bid.

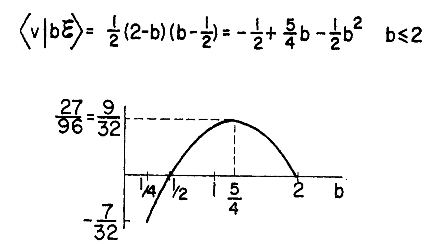

Fig. 2. Expected profit as function of our bid.

distributions are shown in Fig. 1 along with the cumulative and complementary cumulative distributions that they imply. From the symmetry of the density functions we see immediately that our expected cost $\langle p|\mathcal{E}\rangle = \bar{p}$, is just 1/2 and that the expected lowest bid of our competitors $\langle \ell|\mathcal{E}\rangle$ is 1. However, it would be folly to assume that these expected values will occur.

We can now return to (8) and substitute these results,

$$\begin{aligned}\langle v|b\mathcal{E}\rangle &= \int_p\int_\ell \langle v|bp\ell\mathcal{E}\rangle\{p|\mathcal{E}\}\{\ell|\mathcal{E}\}\\ &= \int_p\int_{\ell=b}^{\infty}(b-p)\{p|\mathcal{E}\}\{\ell|\mathcal{E}\}\\ &= \int_p(b-p)\{p|\mathcal{E}\}\int_{\ell=b}^{\infty}\{\ell|\mathcal{E}\}\\ &= \{\ell > b|\mathcal{E}\}\int_p(b-p)\{p|\mathcal{E}\}\\ &= \{\ell > b|\mathcal{E}\}(b - \langle p|\mathcal{E}\rangle)\\ &= \{\ell > b|\mathcal{E}\}(b-\bar{p}). \end{aligned} \tag{10}$$

Our expected profit, given that we bid b, is therefore the product of the probability that our competitor's lowest bid will exceed ours and the difference between our bid and our expected cost. From Fig. 1 we have that our expected profit, given that we bid b, is simply

$$\langle v|b\mathcal{E}\rangle = \frac{1}{2}(2-b)\left(b-\frac{1}{2}\right) \quad 0 \le b \le 2. \tag{11}$$

Figure 2 shows this expected profit as a function of our bid. Note that a maximum profit of 9/32 is achieved by making the bid equal to 5/4. Of course, the same result is obtained by setting the derivative of (11), with respect to b, equal to zero and solving for b. Note that the expected profit is positive for bids in the range between 1/2 and 2, and negative for bids between 0 and 1/2. Formally,

$$\langle v|\mathcal{E}\rangle = \operatorname*{Max}_b \langle v|b\mathcal{E}\rangle = \left\langle v\middle|b=\frac{5}{4},\mathcal{E}\right\rangle = \frac{9}{32} = \frac{27}{96}. \tag{12}$$

We have, therefore, found that the best bidding strategy is to bid 5/4 and that this strategy will have an expected profit of 9/32.

Clairvoyance

Even though we have devised a bidding strategy that is optimum in the face of the uncertainties involved we could still face a bad outcome. We may not get the contract, and if we do get it, we may lose money. It is reasonable, therefore, that if a perfect clairvoyant appeared and offered to eliminate one or both of the uncertainties in the problem, we would be willing to offer him a financial consideration. The question is how large should this financial consideration be.

We shall let C represent clairvoyance and C_x represent clairvoyance about a random variable x. Thus any probability assignment conditional on C_x is conditional on the fact that the value of x will be revealed to us. We define v_{C_x} as the increase in profit that arises from obtaining clairvoyance about x. It is clear that the expected increase in profit owing to clairvoyance about x,$\langle v_{C_x}|\mathcal{E}\rangle$ is just the difference between the expected profit that we shall obtain as a result of the clairvoyance $\langle v|C_x\mathcal{E}\rangle$ and the expected profit that we would obtain without clairvoyance $\langle v|\mathcal{E}\rangle$; thus,

$$\langle v_{C_x}|\mathcal{E}\rangle = \langle v|C_x\mathcal{E}\rangle - \langle v|\mathcal{E}\rangle. \tag{13}$$

We compute the expected profit given clairvoyance about x,$\langle v|C_x\mathcal{E}\rangle$, by evaluating the expected profit given that we know x,$\langle v|x\mathcal{E}\rangle$, for each possible value of x that the clairvoyant might reveal and then summing this expectation with respect to the probability assignment on x,$\{x|\mathcal{E}\}$,

$$\langle v|C_x\mathcal{E}\rangle = \int_x \langle v|x\mathcal{E}\rangle\{x|\mathcal{E}\}. \tag{14}$$

We use the prior probability distribution $\{x|\mathcal{E}\}$ for x in this calculation because up to the moment that the clairvoyant actually reveals the value of x the probability that we must assign to his statement about x is based only on our prior knowledge $\mathcal{E}$.

By using (13) and(14) we can assign the expected value in monetary units of eliminating any uncertainty in the problem. We shall now apply these results to computing the value of eliminating the uncertainty in our cost and in the lowest competitive bid in the problem.

Analysis of Clairvoyance about Our Cost

We now determine $\langle v_{C_p}|\mathcal{E}\rangle$, the expected value to us of eliminating uncertainty about our cost p. From (13) and (14) we write

$$\langle v_{C_p}|\mathcal{E}\rangle = \langle v|C_p\mathcal{E}\rangle - \langle v|\mathcal{E}\rangle \tag{15}$$

$$\langle v|C_p\mathcal{E}\rangle = \int_p \langle v|p\mathcal{E}\rangle\{p|\mathcal{E}\}. \tag{16}$$

The expected profit $\langle v|p\mathcal{E}\rangle$ that we would make if we knew our cost is just the expected profit that we would make if we bid b and if our cost was p maximized with respect to our bid b; that is,

$$\langle v|p\mathcal{E}\rangle = \operatorname*{Max}_b \langle v|bp\mathcal{E}\rangle. \tag{17}$$

Using the assumptions of (6) and (7) we invoke the expansion concept to write

$$\begin{aligned}\langle v|bp\mathcal{E}\rangle &= \int_\ell \langle v|bp\ell\mathcal{E}\rangle \{\ell|\mathcal{E}\} \\ &= \int_{\ell=b}^{\infty}(b-p)\{\ell|\mathcal{E}\} = (b-p)\{\ell>b|\mathcal{E}\} \\ &= (b-p)\frac{1}{2}(2-b)\end{aligned} \tag{18}$$

Note that if we take the expectation of this equation with respect to the prior on p we immediately obtain (11).

We determine our bid by maximizing $\langle v|bp\mathcal{E}\rangle$ with respect to b. By setting its derivative with respect to $b = 0$ we find immediately that

$$\frac{d}{db}\langle v|bp\mathcal{E}\rangle = 0 \rightarrow 2 - b - (b-p) = 0, 2b = 2 + p.$$

$$b = 1 + \frac{p}{2} \tag{19}$$

The bid should be equal to 1 plus 1/2 the value of our cost. When we insert this result in (17) we find that the expected profit given that our cost is p is

$$\langle v|p\mathcal{E}\rangle = \left\langle v|b = \left(1 + \frac{p}{2}\right), p\mathcal{E}\right\rangle = \frac{1}{2}\left(1 - \frac{p}{2}\right)^2. \tag{20}$$

Now we compute the expected profit given clairvoyance about p from (16),

$$\langle v|C_p\mathcal{E}\rangle = \int_p \langle v|p\mathcal{E}\rangle\{p|\mathcal{E}\} = \int_0^1 dp \cdot \frac{1}{2}\left(1 - \frac{p}{2}\right)^2 = \frac{7}{24} = \frac{28}{96}. \tag{21}$$

This expected profit is 28/96, 1/96 more than we obtained in (12) in the case of no clairvoyance. Therefore, following (15) we have found

$$\langle v_{C_p}|\mathcal{E}\rangle = \langle v|C_p\mathcal{E}\rangle - \langle v|\mathcal{E}\rangle = \frac{28}{96} - \frac{27}{96} = \frac{1}{96}. \tag{22}$$

The expected increase in our profit that will result from having our cost revealed to us by a clairvoyant is consequently 1/96.

If the value of this analysis depended on the actual existence of clairvoyance then it would have only theoretical interest. However, since clairvoyance represents complete elimination of the uncertainty about a random variable it follows that what we would pay for clairvoyance should be an upper bound on any experimental program that purports to aid us in eliminating uncertainty about that variable. Thus, in the present problem the company should not hire any cost accounting or production experts to aid in eliminating uncertainty about p unless the cost of these services is considerably below $\langle v_{C_p}|\mathcal{E}\rangle$. The concept of clairvoyance plays the same role in analyzing decision problems that the concept of a Carnot engine plays in analyzing thermodynamic problems. These theoretical constructs provide bench marks against which practical realizations can be tested.

Analysis of Clairvoyance about the Lowest Competitive Bid

Now we determine what it is worth to know the lowest competitive bid. We write immediately,

$$\langle v_{C_\ell}|\mathcal{E}\rangle = \langle v|C_\ell\mathcal{E}\rangle - \langle v|\mathcal{E}\rangle \tag{23}$$

$$\langle v|C_\ell\mathcal{E}\rangle = \int_\ell \langle v|\ell\mathcal{E}\rangle\{\ell|\mathcal{E}\} \tag{24}$$

and

$$\langle v|\ell\mathcal{E}\rangle = \operatorname*{Max}_b \langle v|b\ell\mathcal{E}\rangle. \tag{25}$$

We can perform expansion in terms of our cost p just as we did in (18) in terms of ℓ. We write,

$$\langle v|b\ell\mathcal{E}\rangle = \int_p \langle v|bp\ell\mathcal{E}\rangle\{p|\mathcal{E}\}. \tag{26}$$

From (9) we have immediately that

$$\langle v|b\ell\mathcal{E}\rangle = \begin{cases}\int_p (b-p)\{p|\mathcal{E}\} = b - \bar{p} & \text{if } b < \ell \\ 0 & \text{if } b > \ell.\end{cases} \tag{27}$$

Therefore if $\ell < \bar{p}$ do not bid; if $\ell > \bar{p}$ bid ℓ^-, just below ℓ.

It is easy to see why this is the best bid. Since we know the lowest competitive bid ℓ we can get the contract by bidding just under ℓ. However, if ℓ is already less than our expected cost then we would expect to lose money by such a strategy for it would be better not to get the contract at all. Consequently, if ℓ is less than our expected cost $\bar{p}$ we should not bid; while if ℓ is greater than $\bar{p}$ we should bid just below ℓ. Therefore we have

$$\langle v|\ell\mathcal{E}\rangle = \operatorname*{Max}_b \langle v|b\ell\mathcal{E}\rangle = \begin{cases}\ell - \bar{p} & \text{if } \ell > \bar{p} \\ 0 & \text{if } \ell < \bar{p}.\end{cases} \tag{28}$$

The expected profit given the lowest competitive bid ℓ, $\langle v|\ell\mathcal{E}\rangle$, is plotted as a function of ℓ in Fig. 3. To find the expected profit given clairvoyance about ℓ, $\langle v|C_\ell\mathcal{E}\rangle$, we integrate this function with respect to the ℓ density function of Fig. 1 and obtain

$$\langle v|C_\ell\mathcal{E}\rangle = \int_\ell \langle v|\ell\mathcal{E}\rangle\{\ell|\mathcal{E}\} = \int_{1/2}^2 dl\left(\ell - \frac{1}{2}\right)\cdot\frac{1}{2} = \frac{9}{16} = \frac{54}{96}. \tag{29}$$

The expected profit given clairvoyance about ℓ is 54/96, twice as large as the expected profit of 27/96 obtained in (12) where no clairvoyance was available. The expected

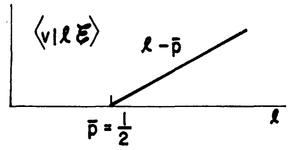

Fig. 3. Our expected profit as function of lowest competitive bid.

increase in profit with clairvoyance about ℓ is, therefore,

$$\langle v_{C_\ell}|\mathcal{E}\rangle = \langle v|C_\ell\mathcal{E}\rangle - \langle v|\mathcal{E}\rangle = \frac{54}{96} - \frac{27}{96} = \frac{27}{96}. \tag{30}$$

By comparing (30) and (22) we see that the expected increase in profit because of clairvoyance about the lowest competitive bid ℓ is 27 times as great as the expected increase in profit because of clairvoyance about our cost p. Yet the density functions for p and ℓ shown in Fig. 1 reveal that the range of uncertainty in ℓ is only twice the range of uncertainty in p. Why, therefore, is information about the lowest competitive bid so much more valuable than information about our cost? The explanation is that we can use information about ℓ far more effectively in controlling the profitability of the situation than we can use information about our costs. If we know ℓ then we can control immediately whether or not we get the contract, although we may still lose money even if we get it. Knowing p prevents us from bidding on any contract that would be unprofitable for us if we got it but is no help in determining whether or not we will get it.

Analysis of Clairvoyance about Our Costs and the Lowest Competitive Bid

Suppose that we are now offered clairvoyance about our cost p and the lowest competitive bid ℓ. What would this joint information be worth? We let $v_{C_{p\ell}}$ represent the increase in profit because of clairvoyance about both p and ℓ. Then in direct analogy with our earlier results for clairvoyance about a single random variable we can write

$$\langle v_{C_{p\ell}}|\mathcal{E}\rangle = \langle v|C_{p\ell}\mathcal{E}\rangle - \langle v|\mathcal{E}\rangle \tag{31}$$

$$\langle v|C_{p\ell}\mathcal{E}\rangle = \int_{p,\ell}\langle v|p\ell\mathcal{E}\rangle\,\{p,\ell|\mathcal{E}\} \tag{32}$$

and

$$\langle v|p\ell\mathcal{E}\rangle = \max_b \langle v|bp\ell\mathcal{E}\rangle$$

$$\langle v|bp\ell\mathcal{E}\rangle = \begin{cases} b - p & \text{if } b < \ell \\ 0 & \text{if } b > \ell. \end{cases} \tag{33}$$

Therefore, if $\ell < p$, do not bid; if $\ell > p$, bid ℓ^-. Since we know both our cost p and the lowest competitive bid ℓ we can get the contract by bidding slightly less than ℓ and we want it if this bid exceeds our cost p. Consequently, our expected profit given p and ℓ will be $\ell - p$ if ℓ exceeds p because we shall bid just less than ℓ on the contract and get it. The expected profit will be zero if ℓ is less than p because we shall not bid, e.g.,

$$\langle v|p\ell\mathcal{E}\rangle = \begin{cases} \ell - p & \text{if } \ell > p \\ 0 & \text{if } \ell < p. \end{cases} \tag{34}$$

Now we substitute the results of (34) and (7) to obtain

$$\begin{aligned}\langle v|C_{p\ell}\mathcal{E}\rangle &= \int_p\int_\ell\langle v|p\ell\mathcal{E}\rangle\{p|\mathcal{E}\}\{\ell|\mathcal{E}\} \\ &= \int_0^1 dp\{p|\mathcal{E}\}\int_{l=p}^2 dl\{\ell|\mathcal{E}\}(\ell - p) \\ &= \frac{1}{2}\int_0^1 dp\int_p^2 d\ell(\ell - p) = \frac{7}{12} = \frac{56}{96}.\end{aligned} \tag{35}$$

The expected profit given clairvoyance about both p and ℓ is therefore 56/96, the highest expected profit we have seen thus far. The expected increase in profit resulting from clairvoyance about p and ℓ is, therefore

$$\langle v_{C_{p\ell}}|\mathcal{E}\rangle = \langle v|C_{p\ell}\mathcal{E}\rangle - \langle v|\mathcal{E}\rangle = \frac{56}{96} - \frac{27}{96} = \frac{29}{96}. \tag{36}$$

The expected value 29/96 of knowing p and ℓ jointly is, therefore, greater than the sum of the 1/96 value of knowing p alone and the 27/96 value of knowing ℓ alone. The further advantage provided by the joint knowledge of p and ℓ is illustrated by the fact that now the company can never lose money whereas in each of the two earlier cases it could.

However, the most striking result of our analysis is the way in which the uncertainty about the lowest competitive bid towers over the uncertainty in our company's cost as a concern of management. In this problem it is worth far more to know what the competition is doing than it is the internal performance of our own company. You might say facetiously that we have demonstrated the dollars and cents payoff in industrial espionage.

Conclusion

The observations made in this paper have wide application. We can treat the case where our clairvoyance is imperfect rather than perfect. The imperfection can be because of a statistical effect arising in nature or the result of incompetence or mendacity or both in the source of the information.

Placing a value on the reduction of uncertainty is the first step in experimental design, for only when we know what it is worth to reduce uncertainty do we have a basis for allocating our resources in experimentation designed to reduce the uncertainty. These remarks are as applicable to the establishment of a research laboratory as they are to the testing of light bulbs.

If information value theory and associated decision theoretic structures do not in the future occupy a large part of the education of engineers, then the engineering profession will find that its traditional role of managing scientific and economic resources for the benefit of man has been forfeited to another profession.

References

[1] R. A. Howard, "Bayesian decision models for system engineering," *IEEE Trans. on Systems Science and Cybernetics*, vol. SSC-1, pp. 36–40, November 1965.

[2] ——, "Dynamic inference," *J. Operations Research*, vol. 13, pp. 712–733, September-October 1965.